

Data Quality Report Template

Phase: Data Collection and Preprocessing Phase

Project Title: Food Consumer Behaviour Analysis

Team ID: PNT2022TMID01234

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Maximum Marks: 3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Quality Assessment & Remediation Table

Data Source	Data Quality Issue	Severity	Resolution Plan
Food Consumer Behavior Dataset	Missing values in 'Monthly Income' and 'Dietary Preference' fields (~4.2% missing records).	Moderate	Applied median imputation for numerical income stratified by occupation and age cohort. Imputed missing dietary preferences with 'Standard / No Restriction'.
Food Consumer Behavior Dataset	Outliers in 'Average Meal Spend' with unrealistic values (\$2,500+ per individual meal due to keystroke entry error).	High	Calculated Interquartile Range (IQR); capped values exceeding Q3 + 1.5*IQR at the 99th percentile threshold (\$180).
Online Delivery Logs	Inconsistent channel naming conventions across records (e.g., 'App', 'mobile_app', 'Delivery App', 'Online').	Moderate	Standardized categorical nomenclature using regex string mapping into three distinct categories: 'Delivery App', 'Dine-In', and 'Takeout'.
Online Delivery Logs	Duplicate customer transaction IDs resulting from retried payment gateway requests (~1.8% duplicates).	High	Implemented deduplication logic in Python using subset=['CustomerID', 'Timestamp', 'Amount'] keeping the first verified record.
Restaurant POS Aggregations	Unformatted text symbols in currency columns (e.g., '\$45.00', 'USD 45', '45.00-').	Low	Stripped non-numeric currency characters using Python string stripping and converted data type to Float64.
Consumer Feedback Logs	Null ratings in 'Delivery Satisfaction Score' for orders cancelled before dispatch (~2.5%).	Low	Separated cancelled orders into distinct fulfillment status cohort and assigned 'N/A - Cancelled' to prevent skewed satisfaction averages.